Problem B

Two Points to the Goal: A Noise-Conditioned Mixture Density Network for Probabilistic Penalty Kick Prediction

Summary

How much can two early positions tell us about where a penalty kick crosses the goal plane? The ball starts at a known point, with observations at 0.05 and 0.10 s. We propagate measurement uncertainty through a flight model with gravity, quadratic drag and a transverse Magnus force. A constant-acceleration solution gives a reference crossing. An adaptive Bayesian sampler and a noise-conditioned mixture density network (MDN) estimate distributions over the crossing point.

At an assumed coordinate noise of 5 mm, the calibrated MDN places the most likely crossing at $(Y, Z) = (2.62, 2.15)$ m. Its joint 50% and 95% prediction regions cover 0.177 and 0.775 m^2, and the probability that the ball centre is inside the goal rectangle, given an airborne crossing, is 97.3%. The Bayesian estimate gives a mode of $(2.650, 2.052)$ m and a conditional probability of 96.3%.

After model selection and calibration, we generated a fresh simulation set with 5,036 airborne crossings. The MDN's median error at 5 mm is 0.290 m. Across four noise levels, its calibrated 50% and 95% regions cover 47.6–49.1% and 94.5–95.2% of endpoints. A $3\,\mathrm{m\,s^{-1}}$ crosswind reduces 50% coverage from 48.7% to 40.9% on matched flights. Tracked penalties and other airborne arcs give case-level checks of the physical model. The coverage figures apply to simulations from the stated launch and measurement assumptions.

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1 Introduction

1.1 Problem Restatement

A penalty kick starts 11 m from a goal that is 7.32 m wide and 2.44 m high [1]. Taking the penalty spot as the origin and the goal line as the plane $x = 11\,\mathrm{m}$, the ball centre begins at $\mathbf{r}(0) = (0, 0, 0.11)\,\mathrm{m}$, and two further positions are observed: $\mathbf{r}(0.05)$ and $\mathbf{r}(0.10)$. The task is to predict the first crossing of the goal plane: the most likely point (Y, Z), joint 50% and 95% prediction regions, and the probability that the ball passes inside the goal. The final crossing is a label for validation only; it may not be used as an input.

Two properties make this inverse problem difficult. First, the trajectory is driven by forces that are not observed. The launch speed is roughly $30\,\mathrm{m\,s^{-1}}$ [6], the Reynolds number is of order 10^5, and the drag coefficient of a football changes by a factor of almost three across the drag crisis [2]. Spin generates a Magnus force whose direction depends on the spin axis, which a tracking system does not measure. Second, the observations are only marginally informative. With the start fixed and two positions observed independently with standard deviation σ, the second difference that estimates acceleration carries noise $\sqrt{5}\,\sigma/\Delta t^2$ (Appendix A). For $\sigma = 5\,\mathrm{mm}$ and $\Delta t = 0.05\,\mathrm{s}$ this is $4.47\,\mathrm{m\,s^{-2}}$, comparable to the aerodynamic accelerations themselves. A predictor that ignores this amplification will be overconfident.

1.2 Background and Approach

The aerodynamics of a football are well studied. Wind-tunnel work gives a subcritical drag coefficient near 0.43, a supercritical value near 0.15, and a critical Reynolds number of 2.2–3.0×10^5 [2, 3]; trajectory analyses of free kicks report effective drag coefficients of 0.25–0.30 [6], and lift coefficients of 0.2–0.3 at moderate spin parameters [4]. More recent work recovers drag and lift curves directly from tracked trajectories [5]. These ranges, rather than a single calibrated value, define our prior. Public tracking datasets make real ball trajectories available: the ISSIA-3D annotations distributed with SoccerNet-v3D provide calibrated, multi-view ball positions for broadcast matches [8], and league-wide spatiotemporal data are now published for the German Bundesliga [9]. Neither source observes spin, and neither marks the instant of foot contact.

We first quantify how uncertainty in the two observed positions propagates into the predicted crossing. We then compare a transparent Bayesian inverse calculation and a fast mixture density network [12]. They share the same forward model and launch prior. Controlled comparisons test which parts of the network help. A separate calibration fold and fresh simulations measure how often the forecast regions contain the true endpoint. Tracked flights provide external checks on the physical assumptions.

2 Materials and Methods

2.1 Assumptions and Approximations

A1. Point mass, rigid sphere. The ball is treated as a point mass with $m = 0.430\,\mathrm{kg}$ and radius $R = 0.110\,\mathrm{m}$, within the IFAB ranges for mass and circumference [1]. The cross-sectional area is $A = \pi R^2 = 0.038013\,\mathrm{m^2}$. Buoyancy is neglected: with a ball density of about $77\,\mathrm{kg\,m^{-3}}$, buoyancy is 1.6% of the weight, smaller than the uncertainty in the aerodynamic coefficients.

A2. Still air. The main model uses zero wind and constant air density $\rho = 1.20\,\mathrm{kg\,m^{-3}}$. A $3\,\mathrm{m\,s^{-1}}$ crosswind is used only as a stress test.

A3. Quadratic drag with one coefficient per flight. The drag coefficient is constant along a trajectory and drawn from $C_D \sim U[0.15, 0.30]$. This range brackets the free-kick estimates of [6] and the supercritical wind-tunnel value of [2]; a wider upper limit of 0.45 is examined as a sensitivity case.

A4. Transverse Magnus force. Only the spin component perpendicular to the relative velocity produces lift. The lift coefficient is $C_L = \min(1.4\,S, 0.30)$ with spin parameter $S = R\,|\boldsymbol{\omega} \times \hat{\mathbf{u}}|/|\mathbf{u}|$. The slope and cap are modelling choices of this study, chosen to match the lift coefficients measured at moderate spin parameters [4, 2]. Zero spin and spin parallel to the velocity produce no lift.

A5. No spin decay. The main model holds spin fixed over a flight. A separate stress test applies exponential decay with a 0.5 s time constant.

A6. Measurement model. The initial position is exact. The two later positions are independent Gaussian observations with a common, known standard deviation $\sigma \in \{2, 5, 10, 15\}$ mm for evaluation. Because only the two later points carry noise, the acceleration uncertainty is $\sqrt{5}\,\sigma/\Delta t^2$; the conservative factor $\sqrt{6}$ applies only if all three positions are noisy (Appendix A).

A7. Plausibility prior. Launch velocity, spin and drag are drawn independently from uniform bounds (Table 1). The bounds are wide plausibility ranges chosen before calibration; they contain the free-kick speeds reported in [6], but they are not a calibrated distribution of real penalty kicks.

Table 1: Prior bounds for the launch parameters (independent uniforms).

Parameter	Lower	Upper
v_x $(\mathrm{m\,s^{-1}})$	18	35
v_y $(\mathrm{m\,s^{-1}})$	-12	12
v_z $(\mathrm{m\,s^{-1}})$	0.5	12
$\omega_x, \omega_y, \omega_z$ $(\mathrm{rad\,s^{-1}})$	-60	60
C_D	0.15	0.30

2.2 Physical Model

Let $\mathbf{u} = \mathbf{v} - \mathbf{w}$ be the ball velocity relative to the air, with $\mathbf{w} = 0$ in the main model. The equations of motion are

$$\dot{\mathbf{r}} = \mathbf{v}, \tag{1}$$

$$\dot{\mathbf{v}} = \mathbf{g} - \frac{\rho A C_D}{2m}\,|\mathbf{u}|\,\mathbf{u} + \frac{\rho A C_L}{2m}\,|\mathbf{u}|^2\,\hat{\mathbf{n}}, \tag{2}$$

where $\mathbf{g} = (0, 0, -9.81)\,\mathrm{m\,s^{-2}}$ and

$$\hat{\mathbf{n}} = \begin{cases} \dfrac{\boldsymbol{\omega} \times \mathbf{u}}{|\boldsymbol{\omega} \times \mathbf{u}|}, & |\boldsymbol{\omega} \times \mathbf{u}| > 0, \\ \mathbf{0}, & \text{otherwise}, \end{cases} \qquad C_L = \min\!\left(1.4\,\frac{R\,|\boldsymbol{\omega} \times \mathbf{u}|}{|\mathbf{u}|^2},\, 0.30\right). \tag{3}$$

In the air-relative frame, the lift term is orthogonal to $\mathbf{u}$ and the drag term dissipates kinetic energy.

The goal plane is crossed when x first reaches 11 m. The crossing time and position are obtained by linear interpolation between the two integration steps that bracket $x = 11\,\text{m}$. A flight ends early if the ball centre falls to $z \leq R$ before reaching the plane; such trajectories are labelled ground hits and are excluded from the endpoint statistics, which are therefore conditional on airborne arrival.

2.3 Numerical Solution and Event Detection

Equations (1)–(3) are integrated with a fixed-step fourth-order Runge–Kutta scheme using $\delta t = 2\,\text{ms}$ and a horizon of 1.5 s. Halving the step to 1 ms changes the interpolated endpoint by less than 0.1 mm on a representative trajectory, well below the millimetre-scale observations. Because the step is fixed, event locations are not exact roots; the linear interpolation error is bounded by the same step-refinement check.

2.4 Bayesian Inference of Launch Conditions

Let $\boldsymbol{\theta} = (\mathbf{v}_0, \boldsymbol{\omega}, C_D)$ and let $\mathbf{y}_1, \mathbf{y}_2$ be the two observed positions. With the start fixed and independent Gaussian errors,

$$p(\mathbf{y}_1, \mathbf{y}_2 \mid \boldsymbol{\theta}) = \prod_{i=1}^{2} \frac{1}{(2\pi\sigma^2)^{3/2}} \exp\left[-\frac{\|\mathbf{y}_i - \mathbf{r}(t_i;\boldsymbol{\theta})\|^2}{2\sigma^2}\right]. \tag{4}$$

The posterior is proportional to the product of (4) and the uniform prior of Table 1. It is sampled with four-stage adaptive importance sampling. Every stage mixes 20% prior draws with a Gaussian proposal. The first Gaussian uses an early-motion estimate, and later Gaussians use the weighted moments of the previous stage. The case study uses 40,000 draws per stage. The paired benchmark starts with 6,000 per stage and retries with 18,000 when conditional ESS falls below 100. The importance weights include the explicit proposal correction $p(\boldsymbol{\theta})/q(\boldsymbol{\theta})$.

The last stage always uses the nominal σ; earlier stages use a widened likelihood to locate the posterior region. Effective sample size (ESS) is reported for every case, and no conclusion is drawn from a sample with low ESS.

From the weighted endpoint sample we estimate the most likely crossing as the maximum of a weighted Gaussian kernel density on a 160×160 grid, and the joint 50% and 95% regions as highest-density regions (HDRs) [14] on the same grid. The grid deliberately extends beyond the goal rectangle so that off-target mass is not truncated.

2.5 Noise-Conditioned Mixture Density Network

Let E denote a forward goal-plane crossing before ground contact. The network estimates $p(Y, Z \mid \mathbf{y}_1, \mathbf{y}_2, \sigma, E)$. The final input has eleven features derived from the two observations, the fixed start and σ. These are three velocity components at the midpoint, three second-difference acceleration components, speed, the signed tangential component and the normal magnitude of the estimated aerodynamic acceleration, a constant-acceleration time to the plane, and σ. Each input is standardised using training data. Apparent drag and lift coefficients and three gravity-shifted duplicates of acceleration were removed from the original sixteen features.

The final network has three hidden layers of 128 units with LayerNorm and GELU. Its output defines $K = 3$ bivariate Gaussian components,

$$p(Y, Z \mid \mathbf{y}_1, \mathbf{y}_2, \sigma, E) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}((Y, Z); \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \, . \tag{5}$$

Each covariance is parameterised by a Cholesky factor with a bounded correlation. The target is the absolute endpoint (Y, Z) and the loss is its negative log-likelihood (NLL). The model uses Adam [7], an initial learning rate of 10^{-3} with cosine decay, batches of 256, and 100 epochs. On each pass through the training set, every trajectory is perturbed anew. Its observation standard deviation is drawn from $U[1, 20]$ mm and shared across the six independently perturbed coordinates. The endpoint label remains unchanged.

2.6 Data Splits, Comparisons and Calibration

We generated 50,000 flights with seed 42. The simulator recorded two early positions, the event type and the endpoint when the ball crossed the plane in flight. There were 41,733 airborne crossings and 8,267 ground hits. Splitting by trajectory before adding noise gave 33,367 airborne training cases, 4,179 validation cases and 4,187 development test cases. We split validation again by trajectory into 2,090 for model selection and 2,089 for calibration. After freezing model and calibration parameters, we generated 6,000 new flights with seed 155107 and evaluated their 5,036 airborne crossings. Within each split and noise level, the methods received the same perturbed observations. All computations use NumPy [16], SciPy [17], PyTorch [18] and Matplotlib [19].

Development comparisons held architecture and data fixed while changing features and targets. Model A maps coordinates and σ to the endpoint. B predicts its residual from a constant-acceleration estimate. C and D use sixteen physics-derived features for absolute and residual targets, respectively. E predicts residuals using coordinates, σ and the baseline endpoint. We also retrained the alternative PI-ResMDN on the canonical data. Final screening compared 16 and 11 features, one and three mixture components, one to four hidden layers with SiLU, then four activations at the selected depth. Each controlled comparison used training seeds 11, 22 and 33. The selection score was the mean validation NLL, equally weighted across the four noise scenarios. When a simpler candidate's mean disadvantage was within one paired seed standard deviation and its mean coverage gap increased by at most 1.5 percentage points, we chose the simpler candidate.

The mixture mean gives the point prediction. We report its median and 90th-percentile Euclidean errors, NLL in physical units, observed coverage and mean area of joint high-density regions, and energy score for paired probabilistic forecasts [15]. For calibration, we fit one standard-deviation multiplier per noise level by NLL on the calibration fold. We then adjusted the 50% and 95% region thresholds using the empirical rank of each calibration endpoint within its forecast density. These threshold adjustments change the reported regions while preserving the fitted density and its mean. We report both pre- and post-calibration coverage to show the effect on new flights.

3 Results

3.1 Answer for the Stated Kick

For the stated observations, the constant-acceleration calculation gives $T = 0.405\,\mathrm{s}$ and $(Y, Z) = (2.495, 1.920)\,\mathrm{m}$. At the representative noise scenario of 5 mm, the calibrated MDN mode is

(2.62, 2.15) m and the mixture mean is (2.70, 1.97) m. The joint regions in Fig. 1 have areas of 0.177 and 0.775 m^2. The probability of the ball centre crossing inside the rectangle is 97.3% conditional on E. The Bayesian mode is (2.650, 2.052) m, with areas of 0.219 and 0.809 m^2 and a conditional probability of 96.3%. Its effective sample size is 2,893. Both estimates use the same stated coordinates and noise assumption.

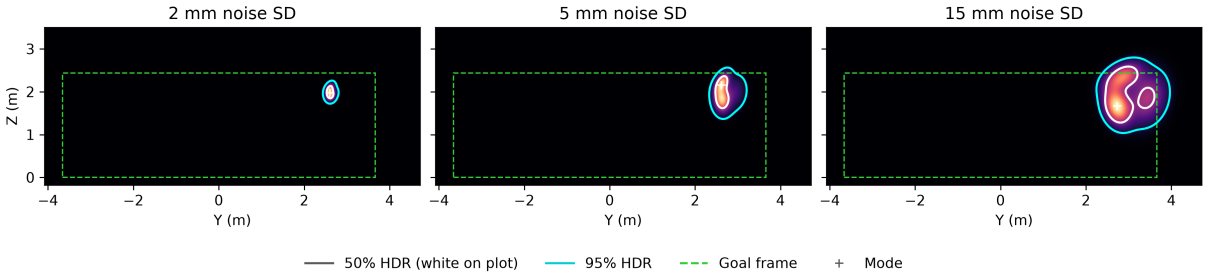


Figure 1: Calibrated MDN crossing densities for the stated kick at 2, 5 and 15 mm observation noise. The white and cyan contours mark the nominal joint 50% and 95% regions. The dashed rectangle is the goal opening. The white cross marks the density mode.

Table 2: MDN answer for the stated kick under three observation-noise scenarios. The final column is the probability of the ball centre being inside the goal rectangle conditional on an airborne crossing. The case uses the preselected model seed 11.

Noise (mm)	Mode Y (m)	Mode Z (m)	50% area (m^2)	95% area (m^2)	In-frame
2	2.60	1.98	0.035	0.151	$> 99.9\%$
5	2.62	2.15	0.177	0.775	97.3%
15	2.73	1.67	0.819	2.370	80.8%

3.2 Feature, Target and Architecture Comparisons

For the same launch parameters, the original alternative simulator and our canonical model produced a 0.240 m RMS endpoint difference on jointly airborne cases. Retraining the alternative on the canonical data made the comparison in Table 3 possible. At 5 mm, the direct physical-feature model C improves on the coordinate model A. Model D is best by NLL at 2 mm, but reaches an NLL of 9.88 at 15 mm on the fresh set. In the development test, roughly 3% of its predictions at that noise level assigned very low density to the true endpoint (NLL above 100). The constant-acceleration calculation can be unstable or have no forward crossing when applied to noisy coordinates. These observations favour direct prediction across the range of measurement scenarios.

The 11-feature version kept performance close to the 16-feature version on validation (NLL -0.0613 and -0.0556). With 11 features and two SiLU layers, using one rather than three Gaussian components changed mean validation NLL from -0.0613 to 0.1205. With SiLU, one to four layers gave scores of 0.0500, -0.0613, -0.0898 and -0.0958. The three-to-four-layer difference (0.0061) was below the paired seed standard deviation (0.0078), so we selected three layers. At that depth, GELU led with -0.1006, followed by SiLU (-0.0898), ReLU (-0.0722) and Tanh (-0.0643). The sequential screen fixes width at 128 and compares activations at the selected depth.

Three further validation comparisons changed one setting at a time. Removing component correlation, using a fixed learning rate and reusing a single noise realisation per trajectory yielded

mean NLL values of −0.0245, 0.0298 and 0.0010, respectively, compared with −0.1006 for the final recipe.

Table 3: Models scored on the same 5,036 fresh airborne crossings at 5 mm. Point forecasts are mixture means. Each value averages three training seeds except the legacy MDN, which uses its original checkpoint. Lower values are better.

Model	Median (m)	90th percentile (m)	NLL
Legacy MDN	0.324	0.675	0.176
A: coordinates, direct	0.309	0.665	0.049
B: coordinates, residual	0.306	0.660	0.120
C: physical features, direct	0.293	0.643	−0.182
D: physical features, residual	0.290	0.648	0.189
E: coordinates and baseline, residual	0.292	0.650	−0.048
PI-ResMDN, canonical retraining	0.433	1.086	1.089
Calibrated final MDN	0.290	0.644	**−0.219**

3.3 Blind-Test Accuracy and Calibration

The final MDN has the lowest NLL among the compared networks at 5, 10 and 15 mm. At 5 mm its median endpoint error is within 3 mm of model C and nearly tied with D. The constant-acceleration estimate has a median error of 0.439 m over the 5,030 cases in which it produces a crossing. It fails on the other six, and on 288 cases at 15 mm.

Table 4: Final MDN on 5,036 independently generated airborne crossings, averaged over three training seeds. The two coverage columns show before and after calibration in percent. The area and NLL are post-calibration values.

Noise (mm)	Median (m)	NLL	50% pre/post	95% pre/post	95% area (m^2)
2	0.130	-1.682	49.9 / 49.1	94.8 / 94.5	0.318
5	0.290	-0.219	48.2 / 47.8	95.6 / 94.9	1.246
10	0.466	0.623	47.2 / 48.8	96.4 / 95.2	2.373
15	0.566	0.956	45.3 / 47.6	96.9 / 94.5	2.804

At 15 mm, the empirical 50% coverage rises from 45.3% to 47.6% after calibration. The 95% coverage moves from 96.9% to 94.5% while the mean area shrinks from 3.162 to 2.804 m^2. The 50% coverage at 5 mm moves slightly away from target. Across the four scenarios, post-calibration 50% coverage ranges from 47.6% to 49.1% and 95% coverage from 94.5% to 95.2% (Fig. 2).

3.4 Paired Comparison with the Physics Posterior

We sampled 32 fresh airborne flights at 5 mm for a paired benchmark. Two Bayesian runs remained below conditional ESS 100 after retry, leaving 30 common cases. The median endpoint errors are 0.379 m for constant acceleration, 0.314 m for the Bayesian sampler, 0.324 m for the legacy MDN and 0.340 m for the final MDN (seed 11). The Bayesian and final MDN median energy scores are 0.210 and 0.211. On the same CPU, their median inference-plus-sampling times are 2.04 s and 1.02 ms. The small, ESS-filtered subset limits the accuracy ranking.

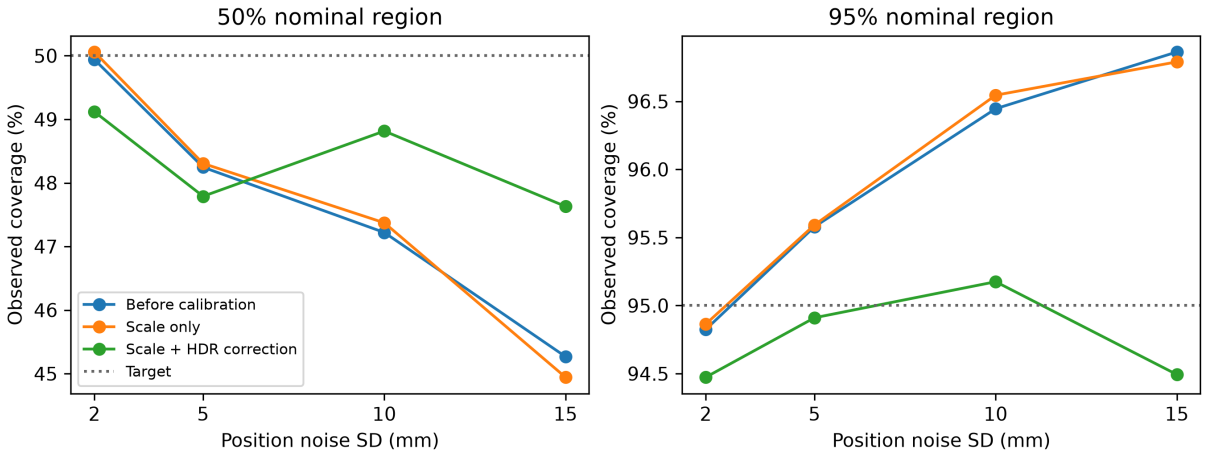


Figure 2: Observed coverage on the independent test set. The dotted lines mark the nominal levels. Per-noise scale adjustment changes little on its own. The density-rank adjustment has the largest effect at higher noise.

3.5 Distribution Shift

For each perturbation we keep the trained network and calibration fixed. We simulate the same initial parameters and compare trajectories that cross in flight under both the nominal and perturbed model. Evaluation uses the same observation perturbations at 5 mm. In a $3\,\mathrm{m\,s^{-1}}$ crosswind (5,001 paired flights), 50% coverage falls from 48.7% to 40.9% and 95% coverage from 94.3% to 92.6%. With exponential spin decay of time constant 0.5 s (5,003 pairs), 50% coverage falls from 48.2% to 36.3% even as median point error falls from 0.293 to 0.263 m. Wind and spin decay change the event label of 70 and 94 flights, respectively, among the 6,000 fresh launch states.

3.6 Sensitivity to the Drag Prior

Because the drag coefficient is the least constrained aerodynamic input, the case was re-run with the upper prior limit raised from 0.30 to 0.45. Three independent seeds were used at each setting, with 40,000 samples per stage (Table 5). The 50% area changes by +1.1% and the 95% area by −0.5%, both smaller than the spread between seeds, and the posterior mean moves by about 2 cm. Within this model, the reported regions are therefore insensitive to the upper drag limit; this does not establish insensitivity to other aerodynamic assumptions.

Table 5: Sensitivity of the case posterior to the upper limit of the drag prior ($\sigma = 5$ mm, three seeds, 40,000 samples per stage). Ranges are minima and maxima over seeds; means are averages over seeds.

C_D prior	50% area (m^2)	95% area (m^2)	Mean (Y, Z) (m)
$U[0.15, 0.30]$	0.213 (0.205–0.219)	0.806 (0.789–0.821)	(2.696, 1.988)
$U[0.15, 0.45]$	0.215 (0.210–0.222)	0.802 (0.792–0.809)	(2.689, 1.971)

3.7 Model-Form Check on Tracked Free-Kick Arcs

The statistical evaluation above is entirely simulation-based. As an external check of the model form, we used two airborne arcs from the ISSIA-3D annotations [8], tracked at 25 fps: frames 608–

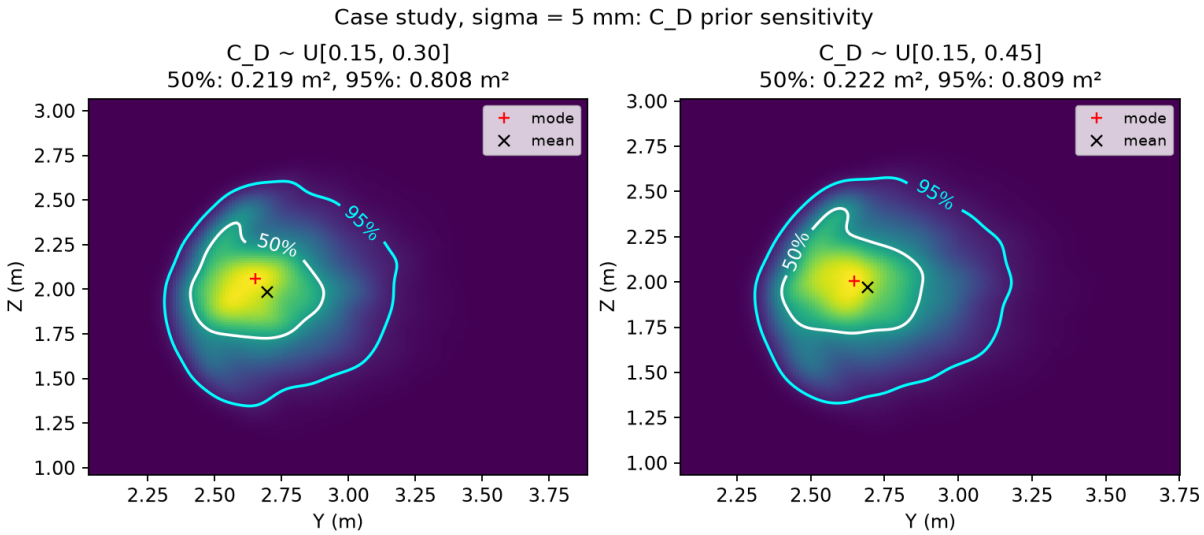


Figure 3: Case posterior at $\sigma = 5$ mm for the main drag prior (left) and the widened prior (right). Contours and areas are from one realisation (seed 8047); the three-seed means and ranges are given in Table 5. The two regions are visually indistinguishable at the resolution of the plot.

622 and 1144–1172, with the first six frames used to fit the initial state and the remaining frames held out. The arcs are free kicks and long passes, not penalties, and they carry centimetre-level reconstruction error; they can test the trajectory model but not the penalty prior.

Table 6 shows the held-out forecast RMSE. Gravity alone is clearly inadequate (0.697 and 2.179 m). Adding quadratic drag with $C_D = 0.22$ reduces the error to 0.489 and 1.546 m; $C_D = 0.45$ gives 0.469 and 1.584 m, so the larger drag is not better on both arcs. A vertical net-lift term fitted on the prefix reduces the error further, to 0.188 and 1.084 m, with fitted $L = 5.76 \pm 4.99$ and $6.89 \pm 3.83\,\mathrm{m\,s^{-2}}$ and fitted $C_D = 0.363 \pm 0.169$ and 0.328 ± 0.167. This is evidence that a vertical force term is worth considering, but the fitted parameters are not identifiable at the precision needed for calibration, and the lift term may absorb tracking or reconstruction error. We therefore use these arcs only as a model-form check, not as a calibration set.

Table 6: Held-out forecast error (RMSE, metres) for two ISSIA-3D airborne arcs. The first six frames of each arc are used to fit the initial state (and, for the last column, drag and vertical lift); later frames are held out. Final error is the error at the last observed frame.

Arc	Quantity	Gravity	$C_D = 0.22$	$C_D = 0.45$	Drag + lift
608–622	Forecast RMSE	0.697	0.489	0.469	**0.188**
	Final error	1.243	0.863	0.811	**0.308**
1144–1172	Forecast RMSE	2.179	1.546	1.584	**1.084**
	Final error	4.169	2.876	2.942	**2.119**

3.8 Prefix Forecasts on Tracked Flights

For two tracked DFL penalties [9], a separate physical inference uses the first three measured positions at the actual frame times and predicts the held-out portion in local coordinates. Table 7 compares drag with zero spin against the same drag prior and unknown spin. The assumed per-coordinate noise is 15 mm and the table averages two seeds. Rotation improves the first penalty's gate estimate and slightly worsens the second's. In two further checks, adding spin

changes held-out RMSE from 0.338 to 0.366 m on an ISSIA segment and from 2.266 to 2.301 m on the two-dimensional reconstruction of the Barber and Carré figure [10] (at its 67 mm sensitivity setting).

Table 7: Two DFL penalties with seven scored positions before each reference crossing. Both trajectory RMSE and crossing-point error are in metres. The crossing forecasts condition on airborne arrival at each observed goal plane.

	Trajectory RMSE		Crossing error	
Event	Zero spin	Unknown spin	Zero spin	Unknown spin
J03WOH	0.347	0.292	0.645	0.543
J03WR9	0.309	0.305	0.542	0.553

Across 11 additional screened DFL match segments, unknown spin lowers held-out RMSE in 7 of 11 segments at each of the 15 and 30 mm assumptions. The paired median improvements are 0.027 and 0.011 m. These segments have no goal-plane endpoint labels, and screening on smooth flight and uncertain contact status limits generalisation. The tracked examples assess physical extrapolation on a case basis. They do not calibrate the MDN.

4 Discussion

The main gain is a better forecast density. At moderate noise, model D, model C and the final MDN often return similar point forecasts, yet the final network assigns more appropriate density to the true endpoint. Coverage and area give complementary checks. In the high-noise setting a calibrated inner region can cover more true endpoints while the outer region becomes smaller. Crosswind and spin decay show that accurate point forecasts can coexist with miscalibrated joint regions.

The Bayesian sampler exposes dependence on launch and aerodynamic assumptions, while the MDN amortises the inverse calculation and is useful for rapid sensitivity studies. Their agreement on the posed case is agreement within a shared physical model and prior. The original plausibility prior generates no launches below $18\,\mathrm{m\,s^{-1}}$ or exactly zero spin. The case forecast also assumes a known measurement noise scale. Real tracking errors, wind and spin are seldom independently known in the available examples.

All network endpoint metrics condition on airborne goal-plane crossing. For arbitrary kicks, the geometric in-frame probability would require multiplying $P(E \mid \mathbf{y}_1, \mathbf{y}_2, \sigma)$ by the conditional in-frame probability. The goal rectangle applies to the centre and ignores the ball radius in the in-frame criterion, contact with the frame and goalkeeper intervention. These distinctions separate a geometric forecast from a match scoring probability. The limited DFL, ISSIA and figure-read cases cannot establish real-penalty coverage. The PFF World Cup release [11] is a possible source for further case audits.

5 Conclusion and Outlook

Two early observations and a known start support a probabilistic forecast of the crossing location. At 5 mm, the calibrated MDN answers the posed case with mode $(2.62, 2.15)$ m, joint areas 0.177 and $0.775\,\mathrm{m}^2$, and a 97.3% conditional in-frame probability. The final MDN has a median error of 0.290 m on the independent 5 mm test. Calibrated coverage is close to nominal under the simulation assumptions and degrades under physical shifts.

The next step is to collect high-rate, three-dimensional penalty tracks with independently characterised timing and tracking error. A sealed evaluation set should include misses, low-speed and low-spin shots, and early ground contact. A separate event model could then predict airborne crossing before the conditional location forecast is applied.

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A Propagation of Position Noise to Acceleration

Let the start $\mathbf{r}_0$ be exact and let the two observations be

$$\mathbf{r}_1 = \mathbf{r}_0 + \mathbf{v}\Delta t + \tfrac{1}{2}\mathbf{a}\Delta t^2 + \boldsymbol{\varepsilon}_1, \qquad \mathbf{r}_2 = \mathbf{r}_0 + 2\mathbf{v}\Delta t + 2\mathbf{a}\Delta t^2 + \boldsymbol{\varepsilon}_2, \tag{6}$$

with independent errors of variance σ^2 per component. Eliminating $\mathbf{v}$ gives the second difference

$$\mathbf{r}_2 - 2\mathbf{r}_1 + \mathbf{r}_0 = \mathbf{a}\Delta t^2 + \boldsymbol{\varepsilon}_2 - 2\boldsymbol{\varepsilon}_1, \tag{7}$$

whose component variance is $(1 + 4)\sigma^2 = 5\sigma^2$. Hence

$$\sigma_a = \frac{\sqrt{5}\,\sigma}{\Delta t^2}. \tag{8}$$

If the start is also noisy, the third error enters with unit weight and the variance becomes $6\sigma^2$, giving the conservative factor $\sqrt{6}$. At $\sigma = 5\,\mathrm{mm}$ and $\Delta t = 0.05\,\mathrm{s}$, $\sigma_a = 4.47\,\mathrm{m\,s^{-2}}$ for a fixed start. For a fixed forecast time near $0.4\,\mathrm{s}$, the acceleration term alone corresponds to a displacement scale of several tenths of a metre. The full crossing uncertainty also involves velocity errors and uncertainty in crossing time. This calculation motivates propagating the observation model through to the endpoint [13].